BRIEF REPORT

Acquired Resistance to Crizotinib in NSCLC with *MET* Exon 14 Skipping

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ABSTRACT

Introduction: *MET* proto-oncogene, receptor tyrosine kinase gene (*MET*) exon 14 skipping is a targetable alteration in lung cancer. Treatment with MET proto-oncogene, receptor tyrosine kinase inhibitor can cause dramatic responses in patients whose cancers have *MET* exon 14 skipping. Little is known, however, about acquired resistance in patients with *MET* exon 14 skipping.

Methods: Biopsy specimens obtained at baseline and at the time of progression for a patient being treated with crizotinib were compared using targeted next-generation sequencing to assess for mechanisms of resistance.

Results: An acquired mutation in the *MET* kinase domain, *D1228N,* was found at time of progression on crizotinib in a patient with *MET* exon 14 skipping.

Conclusions: One potential mechanism of acquired resistance to crizotinib in patients with *MET* exon 14 skipping is through second-site mutations in the *MET* gene. Understanding mechanisms of resistance will be important in optimizing therapy in these patients.

Keywords: NSCLC; *MET* exon 14 skipping; Acquired resistance; Targeted therapy

Introduction

MET proto-oncogene, receptor tyrosine kinase gene (*MET*) exon 14 skipping is reported in approximately 3% to 6% of NSCLC and appears mutually exclusive of other driver alterations.[1–6] A variety of different genomic changes, many of which occur in the intronic segments of *MET*, lead to an alternatively spliced transcript of *MET* in which exon 14 is skipped; the resultant deletion in the juxtamembrane domain results in loss of Cbl E3-ligase binding, leading to decreased ubiquitination and delayed downregulation of proto-oncogene, receptor tyrosine kinase (MET).[1–4] *MET* exon skipping has been observed in both adenocarcinoma and squamous cell carcinoma, and a high prevalence has been reported in sarcomatoid lung carcinoma.[1–7]

Cancers whose tumors have *MET* exon 14 skipping can have dramatic responses to MET inhibitors, inciting hope that this will become another therapeutic target in NSCLC.[3–7] Multiple clinical trials testing the activity of MET inhibitors in this setting are now under way. One early report on the MET inhibitor crizotinib in patients

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with cancers harboring *MET* exon 14 skipping showed a confirmed response rate in eight of 18 of patients (44% [95% confidence interval: 22–69]). An additional five patients (28%) had unconfirmed responses, and the median progression-free survival was not calculable.[8]

Despite remarkable initial responses, however, the disease of patients receiving targeted therapy will invariably progress, and understanding the mechanisms of acquired resistance is critical to developing optimal therapies.[9] The experience with epidermal growth factor receptor (*EGFR*) mutations and anaplastic lymphoma kinase (*ALK*) and ROS1 proto-oncogene receptor tyrosine kinase (*ROS1*) translations in NSCLC has demonstrated that multiple mechanisms of acquired resistance to targeted therapy exist: these include on-target alteration, such as gene amplification and second-site mutations, activation of bypass tracts, and histologic transformation, among others.[10] As yet, little is known about mechanisms of resistance to MET therapy among patients with *MET* exon 14 skipping.

Methods

Patients receiving targeted therapies at Massachusetts General Hospital undergo repeat biopsy at time of progression and are enrolled in an institutional review board–approved protocol of tissue collection and research. A patient who was receiving crizotinib therapy for metastatic lung cancer with *MET* exon 14 skipping underwent a biopsy at time of progression. Tumor genotyping on the baseline and postprogression biopsy samples was performed with the Snapshot next-generation sequencing version 1 assay, which uses anchored multiplex polymerase chain reaction for the detection of single nucleotide variants and insertion/deletions in the genomic DNA of 39 cancer-related genes.[10] Immunohistochemical testing was performed on 5-μm sections of formalin-fixed paraffin-embedded tumor tissue. Total MET staining was performed using the CONFIRM Anti-Total c-MET (SP44) rabbit monoclonal primary antibody (Ventana Medical Systems, Tucson, AZ), and phospho-MET staining was performed using the Anti-MET phospho Y1349 rabbit monoclonal primary antibody (EP2367Y; ab68141) (Abcam, Cambridge, MA) diluted 1:100 in Leica Primary Antibody Diluent (Leica Biosystems Richmond, Inc., Richmond, IL). Targets were retrieved using Epitope Retrieval Solution 2 for 20 minutes, and staining was performed on the Leica BOND RX autostainer using the Polymer-Refine Detection Kit.

Results

Squamous cell cancer of the right lung was diagnosed in a 76-year-old woman with a remote 12–pack-year smoking history in May 2013. She underwent chemotherapy with carboplatin and paclitaxel combined with concurrent radiation for stage IIIA (T2aN2M0) disease. In June 2014, surveillance scans showed multiple bony and liver lesions consistent with metastases. Analysis of a biopsy specimen of a liver lesion confirmed metastatic squamous cell lung cancer. She was treated with palliative radiation to several bony sites of disease, and subsequent treatments for metastatic disease included a clinical trial of an immunotherapy checkpoint inhibitor followed by gemcitabine.

The patient had signed an institutional consent form for clinical tumor genotyping, as well as institutional review board–approved consent forms for tissue collection and research. Tumor genotyping revealed the presence of a *MET* exon 14 skipping mutation, *D1010H*, in her baseline biopsy sample. There was no coexisting driver alteration, including *EGFR* or *KRAS* mutation, or *ALK* or *ROS1* translocation. An atypical *ROS1* mutation (*L2035F*) of uncertain significance and a tumor protein p53 gene (*TP53*) *R158H* mutation were also present at baseline. Because of the presence of the *MET* exon 14 skipping mutation, she was enrolled in a clinical trial of crizotinib [NCT00585195] in August 2015. She had a partial response according to the Response Evaluation Criteria in Solid Tumors, version 1.0, criteria, with a marked decrease in multiple liver metastases. According to restaging scans, her disease ultimately progressed in April 2016, with development of new liver metastases (Fig. 1). A biopsy of one of the new liver lesions was performed, and tumor genotyping with the same platform revealed a new *D1228N* mutation in exon 19 of *MET* in addition to the original exon 14 skipping *D1010H* mutation (Fig. 2). Residue 1228 is located within the activation loop of the MET kinase domain. MET or EGFR amplification was not seen. Again, no other driver alterations were found, and the preexisting *TP53 R158H* and *ROS1 L2035F* mutations were again seen. Immunohistochemical testing of the postprogression biopsy sample confirmed high levels of phospho-MET protein, suggesting activation of MET (see Fig. 2).

Discussion

We have described a patient with NSCLC and *MET* exon 14 skipping in whom acquired resistance to crizotinib developed. Biopsy samples obtained at the time of progression showed an acquired mutation in the *MET* kinase domain, *D1228N*, in addition to the original *D1010H MET* splice site mutation.

Preclinical models in MET-dependent cell lines have demonstrated various mechanisms of resistance to MET inhibitors that can develop. Mutations in *MET*, including at positions *Y1230* and *D1228*, are among the reported

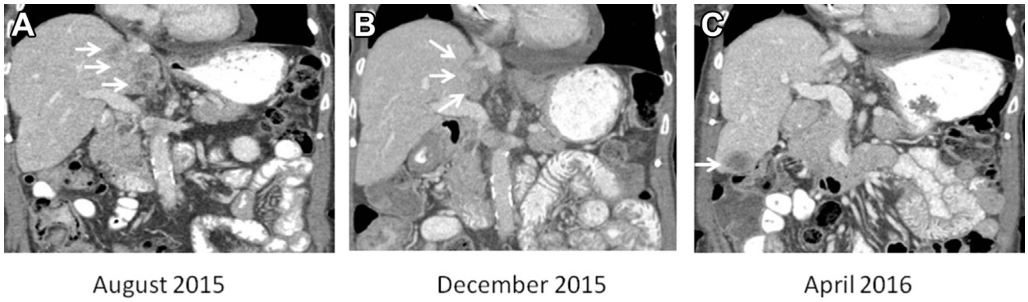

Figure 1. Computed tomography (CT) scans of a patient with *MET* proto-oncogene, receptor tyrosine kinase gene (*MET*) exon 14 skipping showing response and subsequent progression while the patient was receiving crizotinib. (*A*) Contrast-enhanced coronal CT scan before crizotinib. White arrows indicate hepatic metastases. (*B*) Contrast-enhanced coronal CT scan at best response to crizotinib. White arrows indicate interval decrease in size of hepatic metastases. (*C*) Contrast-enhanced coronal CT scan at time of progressive disease. White arrow indicates a new liver metastasis on which a biopsy was performed.

mechanisms of acquired resistance in these models.[11,12] These mutations are already known to be activating in papillary renal cell carcinomas and other cancers.[13] In addition, crystal structures of type I MET inhibitors bound to the MET kinase domain show an important binding interaction with the Y1230 residue, with D1228 playing a role in stabilizing the conformation of the activation loop.[11,12] Therefore, in addition to intrinsic transforming activity, disruption of the drug binding interaction between type I inhibitors and the MET kinase domain is hypothesized to underlie the mechanism of resistance of these specific mutations.

Interestingly, type II inhibitors, which bind to a DFG-out conformation of MET, have less reliance on this

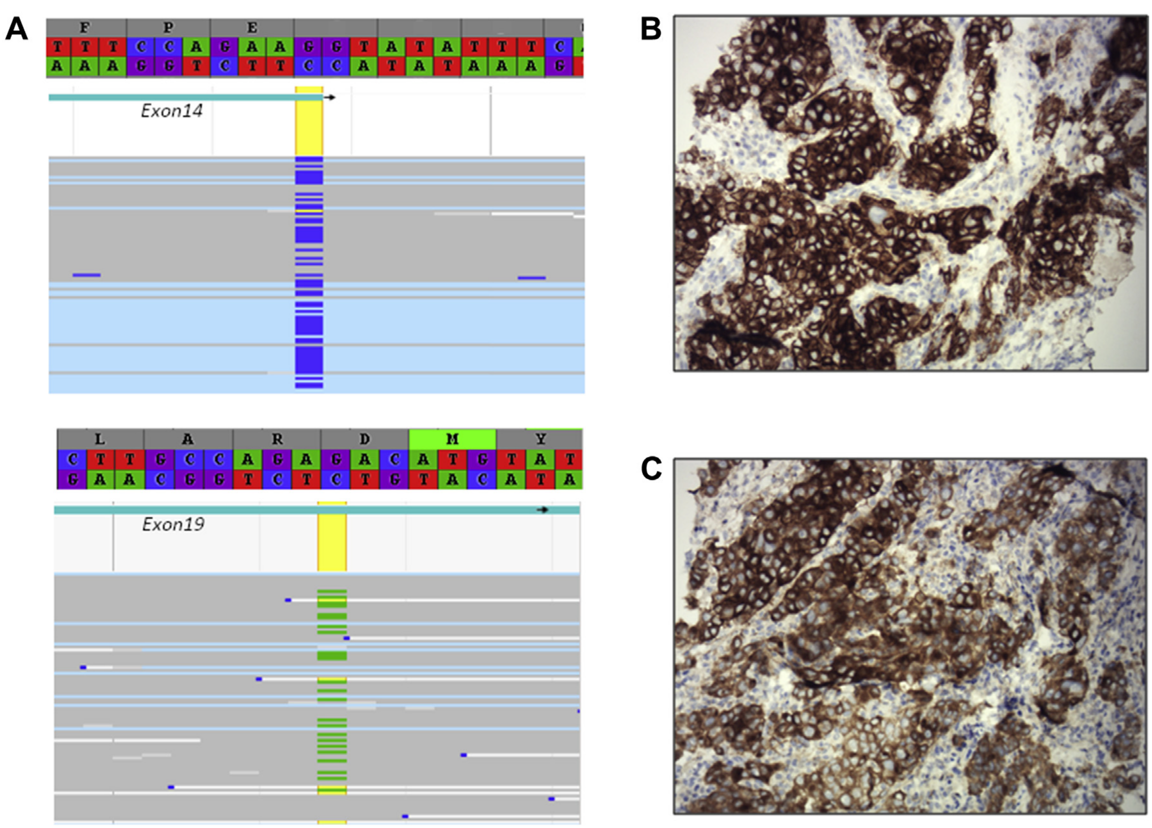

Figure 2. Emergence of a secondary *MET* proto-oncogene, receptor tyrosine kinase gene (*MET*) *D1228N* mutation and proto-oncogene, receptor tyrosine kinase (MET) protein activation upon development of acquired resistance to crizotinib in the setting of *MET* exon 14 skipping. (*A*) Sequence alignment shows the original *MET* exon 14 skipping *D1010H* mutation and the emergence of the secondary *MET D1228N* mutation. Immunohistochemical testing identified high levels of total MET protein (*B*), as well as high levels of active MET protein through immunohistochemical testing for phospho-MET Y1349 (*C*).

particular interaction.[14] Indeed, individual MET inhibitors are known to have differential activity, based on binding mode, against the activating *MET* mutations found in papillary renal cell carcinoma.[15] In cell proliferation assays, acquired mutations at *Y1230* and *D1228* can impair the activity of type I, but not type II, inhibitors.[11,12] These studies suggest the potential for rationally choosing subsequent therapy on the basis of knowledge of the specific acquired mutation and the binding properties of the various clinically available MET inhibitors.

The full spectrum of mechanisms of resistance to MET therapy in *MET* exon 14 skipping remains to be elucidated. Activation of the EGFR pathway due to increased expression of tranforming growth factor-α has also been observed in the preclinical models already described.[11] In addition, modeling of resistance in cell lines to a type II, rather than type I, MET inhibitor showed both a lower prevalence of kinase domain mutations as the mechanism of resistance as well as a different complement of specific mutations among those that were found[12]; the alternative resistance pathways were not further explored in that particular study.

A cancer's ability to develop resistance mechanisms is wide-ranging, and it is unlikely that second-site mutations in the target gene will be the sole mechanism of resistance that is found in patients with *MET* exon 14 skipping. As already demonstrated preclinically, a range of resistance mechanisms, including bypass tract signaling and second-site mutations, can develop in response to a single MET inhibitor.[11] Furthermore, structurally dissimilar MET inhibitors are likely to yield a different propensity for resistance mechanisms. Although preclinical studies have shown that *D1228* and *Y1230* mutations can be mechanisms of resistance to MET inhibitors, to our knowledge, this is the first clinical report of such a mutation arising in a patient with *MET* exon 14 skipping. Repeat biopsy at progression will be critical to defining the spectrum of acquired resistance mechanisms that develop to the numerous MET inhibitors that are currently in clinical trials. The range of resistance mechanisms observed will likely vary depending on intrinsic tumor factors, as well as on the structure, potency, and selective pressure of each individual MET inhibitor.

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